Does Intermunicipal Cooperation Decrease the Property Tax Gap?

Evidence from Italy

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Paper in a Nutshell

Local governments face a fundamental trade-off: political representation favors small jurisdictions BUT economies of scale in complex functions favor large ones.

$\downarrow$

Some small municipalities resolve it by joining forces — pooling staff and expertise without giving up independence. That is **Intermunicipal Cooperation (IMC)**.

$\downarrow$

We ask: does IMC help municipalities reduce tax evasion and improve collection of what they are owed?

$\downarrow$

Short answer: **yes**, but only where enforcement capacity is structurally insufficient.

Motivation & Contribution

The IMC literature focuses more on expenditure:

- Cost savings in waste, transport, public services — mixed results (Bel & Warner, 2015, 2016)
- Quality improvements emerging (Gori et al., 2025)

The revenue side is (almost) neglected:

- Small municipalities have **statutory** taxing power ... but often lack **enforcement capacity**
- Notsu et al. (2025) — IMC improves tax debt recovery in Japan. No *causal* evidence on evasion.

This paper

Evidence that IMC affects **revenue enforcement** — using novel official data on Italy's property tax gap.

Key Definitions

Outcome, institutional setting, and treatment

Our focus: The Italian Property Tax (IMU)

IMU is the main municipal tax on real estate, and the most important source of municipal revenue.

What is taxed: second homes, business premises, rented dwellings, luxury primary residences, non-exempt agricultural land (primary residence **exempt**).

Rate structure: Standard rate 8.6‰ set nationally; municipalities adjust within predefined bands → some degree of statutory **autonomy**

The enforcement problem

The cadastral register determines the taxable base — but municipalities **lack the instruments (personnel, technology, information) to verify that properties are correctly declared — owners may misreport or conceal them — and, in a second step, to audit non-compliant owners**. ⇒ **statutory power ≠ effective collection**

Outcome variable: The IMU Tax Gap

First, let's define what the Tax Gap is

$$\textit{Tax Gap} = \underbrace{(\overset{\substack{\textit{unobservable}\\ \downarrow}}{\textit{Theoretical}} - \textit{Assessed})}_{\text{Assessment Gap (Evasion)}} + \underbrace{(\textit{Assessed} - \textit{Collected})}_{\text{Collection Gap (Residuals)}}$$

We do not compute the IMU Tax Gap ourselves; we rely on Ministry of Economy and Finance (MEF) data that uses estimates of cadastral values of property across all of Italy.

$$\textit{IMU Tax Gap}_i = 1 - \frac{\text{Standardized Revenue}}{\text{Cadastral Value} \times \text{Standard Rate}}$$

Both components reflect the municipality's **inability to enforce its statutory rights**. MEF computes this via cadastral cross-checks unavailable to small municipalities.

Inter-municipal Cooperation in Italy

IMC was designed to nudge small municipalities to cooperate on public services. The main difference from a merger between two or more municipalities is that, in IMC, the participating municipalities only delegate specific services to a joint management body.

$\Rightarrow$ pools enforcement capacity while retaining **political autonomy and separate electoral accountability**

In Italy, two major legal instruments exist: a lightweight contractual agreement (*Convenzione*) and a permanent joint body (*Unione di Comuni*).

The process has been incentivized by the central government with a series of financial incentives and a national law that made IMC mandatory for municipalities below 5,000 inhabitants — but this obligation faced significant implementation challenges.

▸ Legal framework details

Treatment Definition

IMC can jointly manage any municipal function: Registry, Nurseries, Education, Police, Waste Management, Social Services, Public Land, Transportation, Traffic Management, Taxation, Technical Office

Treated = municipality in an IMC covering **Registry + Taxation + Technical Office**

enforcement chain: cadastral records → tax assessment → technical verification

Not-Treated = municipality not cooperating on Registry/Taxation/Technical Office, or not cooperating at all

Data & Descriptives

Sources, sample, and coverage

Data Sources

- **SOSE** — Municipal organizational structure, 2013–2022 (missing: 2014, 2020)
- **MEF** — Official IMU tax gap estimates, 2017–2022 (missing: 2020)
- **ISTAT** — Socio-demographic and fiscal controls

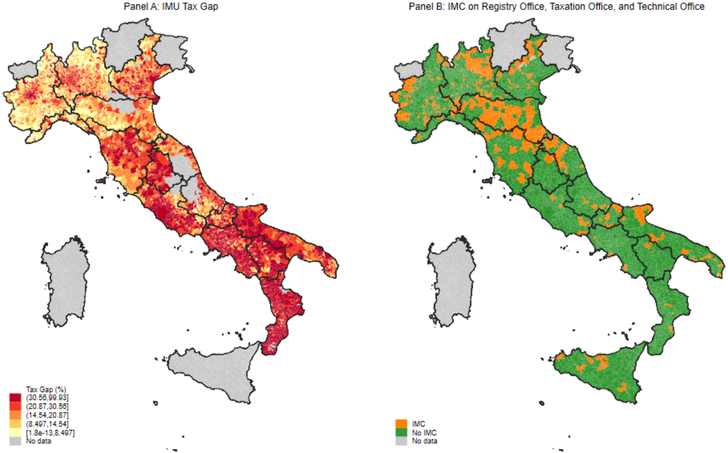
Panel: 5,062 municipalities $\times$ 5 outcome years (2017, 2018, 2019, 2021, 2022)

Descriptive Statistics: Treated vs. Not-Treated (2019)

	Not-Treated		Treated	
	Mean	SD	Mean	SD
IMU Tax Gap	0.222	0.143	0.147	0.106
North (=1)	0.534	0.499	0.753	0.432
Urbanization (1=rural, 3=urban)	2.530	0.594	2.681	0.478
Population	9,900	57,830	4,402	7,467
Population density (per km^2)	399.3	806.8	199.8	332.1
Avg. total income (EUR)	17,714	3,840	18,374	3,187
Avg. employee income (EUR)	19,314	3,991	20,164	3,184
Schools per 1,000 inhab.	1.258	1.023	1.249	1.178
Social expenditure p.c. (EUR)	72.3	49.7	80.2	46.3
Mayor's age	49.4	7.1	49.7	7.3
Observations	3,948		1,114	

Source: SOSE, MEF. 2019 cross-section.

Geographic Distribution



Left: Tax gap (darker = higher). *Right*: Treated (orange) vs. Not-Treated (green). Year 2019.

Empirical Strategy

Identification of the ATT

Empirical Strategy: Motivation

Data problem: IMC started in 2000; the biggest wave of adoption happened between 2013 and 2016. We have data for the outcome only from 2017, thus **we do not observe the outcome before treatment for the majority of IMC municipalities**.

Selection problem: municipalities self-select into IMC based on capacity, geography, and institutions $\Rightarrow$ naive OLS is **biased**.

Solution: Matching with IPW

We identify the ATT via **selection-on-observables** in a cross-sectional setup (year-by-year): we reweight the control group to match the treated covariate distribution.

Our main results are computed on the **2019** cross-section, but we check stability across all years.

Step 1 — Propensity Score via LASSO

We estimate the **propensity score** $p(X_i) = P(\text{Treated}_i = 1 \mid X_i)$ via **LASSO logistic regression** using pre-treatment covariates (2013).

We exclude municipalities that adopted IMC before 2013 (*always treated*): their pre-treatment baseline is unobservable. Only **post-2013 cohort**.

Candidate pool: 38 variables **Retained:** 21 variables

Institutional:	MAQI pillars I–III (bureaucracy, politicians, fiscal performance)
Geographic:	urbanization, altitude zone, coastal, island
Demographic:	pop. size classes, log pop., density
Fiscal/income:	tax burden, income surtax, avg. income, employee income

Regional FE **forced in**

Common support result

Min-Max trimming of the propensity score distribution $\Rightarrow$ final sample $N = 4{,}149$.

Common support Covariate balance

Step 2 — IPW-WLS Estimation

$$\textit{IMU Tax Gap}_i = \beta_0 + \beta_1 \,\text{North}_i + \underbrace{\beta_2 \,(\text{North} \times \text{Treated})_i}_{\text{ATT, North}} + \underbrace{\beta_3 \,(\text{CS} \times \text{Treated})_i}_{\text{ATT, Center-South}} + \gamma' X_i + \varepsilon_i$$

- North_i: 1 if the municipality is in the North, 0 if in the Center-South
- North $\times$ Treated: 1 if a treated municipality is in the North, 0 otherwise
- CS $\times$ Treated: 1 if a treated municipality is in the Center-South, 0 otherwise
- X_i: population, density, avg. income, employee income, schools p.c., social expenditure p.c., mayor's age

IPW weights

Treated: weight $= 1$ (represent themselves) **Controls**: weight $\propto \hat{p}/(1-\hat{p})$
This reweights the control group to match the treated covariate distribution.

Results

Main Results

	Outcome: IMU Tax Gap			
	(2) OLS	(3) OLS+Cov	(4) IPW	(5) IPW+Cov
Treated × North	−0.033***	−0.025***	0.004	−0.005
	(0.004)	(0.004)	(0.005)	(0.005)
Treated × Center-South	−0.076***	−0.062***	−0.080***	−0.063***
	(0.008)	(0.008)	(0.010)	(0.010)
N	5,062	5,062	4,149	4,149
R^2	.288	.356	.334	.402

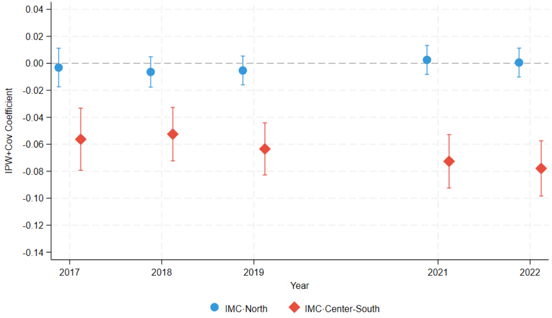
SEs clustered at municipality level. *** $p < 0.01$. (year=2019).

North: OLS+Cov shows −2.5 pp $\rightarrow$ IPW+Cov ≈ 0. Selection, not effect. Under IPW, the North shows no average enforcement gain from IMC.

Center-South: −6.3 pp*** survives full IPW. IMC acts as a **catch-up mechanism** where enforcement capacity is structurally low.

Also robust to Conley spatial SEs and alternative matching estimators (PSM, CEM): ▸ Conley SEs
▸ Matching estimators

Year-by-Year Results



IPW + Cov coefficients by year (2017–2022, excl. 2020).
$N = 4{,}149$. 95% CIs.

- North: **zero** in every year
- CS: **growing** — 2017: -5.6 pp $\rightarrow$ 2022: -7.8 pp
- Consistent with joint tax offices **maturing** over time
- 2019 cross-section is representative

Mechanism: Taxation Office vs. Other-Function Cooperation

Direct test: Column 1 compares Taxation Office cooperators to pure non-cooperators. Column 2 compares *other-function* cooperators (waste, social services, roads — no tax office) to the same pure non-cooperators.

	Outcome: IMU Tax Gap (IPW+Cov)	
	(1) Taxation Office	(2) Other functions
Treated× North	0.014	−0.010
	(0.010)	(0.010)
Treated× CS	−0.059***	−0.018
	(0.019)	(0.027)
N	3,621	3,499

244 Taxation-Office-treated units; SEs clustered.

- Taxation Office: −5.9 **pp*** ** in CS, **zero** in North — confirmed channel
- Other functions: **null in both regions** — rules out generic administrative integration
- Prediction confirmed: the effect is specific to the Taxation Office

Robustness grid

Robustness: Spatial First Difference (SFD)

Why? IPW controls for observables. Spatially persistent unobservables — geography, institutions, local culture — may still confound the estimates.
(Duranton et al., 2011; Klein & Tchuente, 2023)

The idea: difference each municipality against the inverse-distance-weighted (IDW) average of its $k = 4$ nearest *untreated* neighbors:

$$\Delta \mathrm{gap}_i = \mathrm{gap}_i - \bar{\mathrm{gap}}_i^{\mathrm{IDW}}, \qquad w_{ij} \propto 1/d_{ij}$$

Removes unobservables shared with the nearest untreated neighbors within a ∼20–50 km radius.

Estimand: local neighborhood effect — expected to be *smaller* than the IPW ATT (different geographic scale).

Sample: restricted to the *Taxation Office* treatment, against a control pool of pure non-cooperators.

Robustness: SFD Results

	Taxation Office Treatment (pure non-cooperator control pool)	
Δ Tax Gap	(1) Interact	(2) +Cov
Tax Office × North	−0.014***	−0.012**
	(0.005)	(0.005)
Tax Office × CS	−0.035**	−0.032**
	(0.015)	(0.015)
N	4,300	4,300
Covariates		✓

SEs clustered at municipality level. *** $p<0.01$, ** $p<0.05$, * $p<0.1$. Source: SOSE, MEF (2019).

Counterfactual Policy Simulation

If all untreated CS municipalities adopted IMC (≈1,500 potential candidates):

$$\text{Savings}_{i,t} = (-\hat{\beta}_{CS,t}) \times \text{Theoretical Tax Base}_{i,t}$$

Year	Savings (EUR M)	% of theor. rev.
2017	271.5	5.3
2018	234.7	5.0
2019	280.5	5.9
2021	325.8	6.9
2022	350.2	7.4
Total	1,462.7	

2020 excluded (COVID anomaly).

- Range: **235 – 350 M EUR/year** | 5-year total: ≈**1.46 billion EUR**
- Savings grow as joint offices mature over time

Conclusion

1. IMC reduces the IMU gap by -6.3 **pp** in the Center-South; **zero** in the North
2. Effect survives IPW, mechanism test, SFD, and year-by-year checks
3. Driven specifically by the **joint Taxation Office** — not generic cooperation
4. Effect grows over time: joint offices mature, enforcement improves

Policy: Shift from mandatory to **structurally targeted** incentives. Focus on Center-South. Potential recovery: **235–350 M EUR/year**.

IMC is a **state-capacity-building** mechanism. Cooperation removes the enforcement constraint *where it binds most.*

Thank you

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Appendix

Legal Framework: Details

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***Convenzione* (Art. 30):**

- Contractual agreement
- No new legal entity
- Function-specific, renewable

***Unione di Comuni* (Art. 32):**

- Autonomous public body
- Own budget and statutory organs
- Permanent joint management

Regulatory timeline: 2000: voluntary $\rightarrow$ 2010: mandatory ($<$5,000 inhab.) $\rightarrow$ 2024: voluntary again (incentives retained)

Covariate Balance: Love Plot

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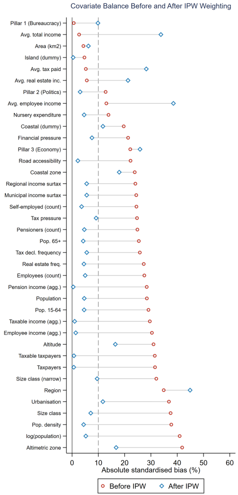

Standardized mean differences before (circles) and after (triangles) IPW reweighting for all 38 candidate covariates.

Common Support Area

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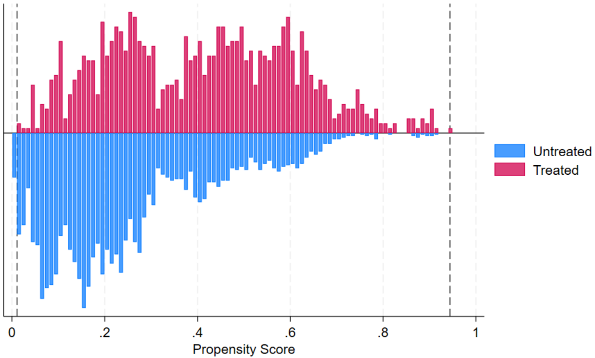

Propensity score distribution for treated (red) and control (blue). Min-Max trimming applied; final sample $N = 4{,}149$.

Mechanism Robustness: 13-Specification Grid

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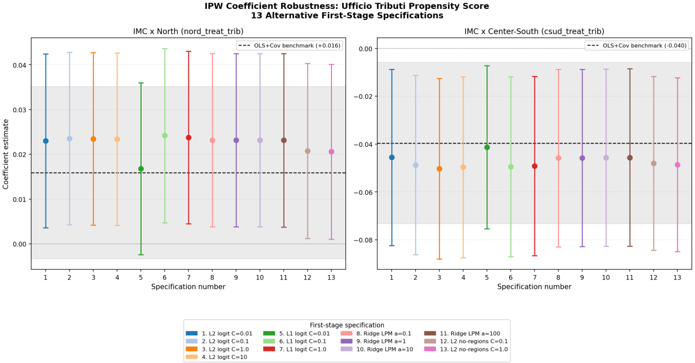

13 alternative first-stage specifications: L1/L2 logit, ridge LPM, $\pm$ regional FE. CS coefficient: -4 to -5 pp** across all 13; North: near zero n.s. throughout. Mechanism result is insensitive to propensity score estimation method.

Robustness: Conley Spatial Standard Errors

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Concern: clustering at the municipality level may understate spatial autocorrelation — nearby municipalities can share unobserved shocks.

Fix: re-estimate the OLS+Cov specification (Column 3 of the main results) with Conley (1999) spatial HAC standard errors at 50km and 100km cutoffs.

	Outcome: IMU Tax Gap (OLS+Cov)	
	Conley 50km	Conley 100km
Treated × North	−0.025***	−0.025***
	(0.008)	(0.008)
Treated × CS	−0.062***	−0.062***
	(0.017)	(0.023)
N	5,062	5,062

Point estimates identical to Column 3; only SEs change.

Point estimate unchanged, still significant at 1% — the Center-South result is not an artifact of spatial autocorrelation.

Robustness: Alternative Matching Estimators

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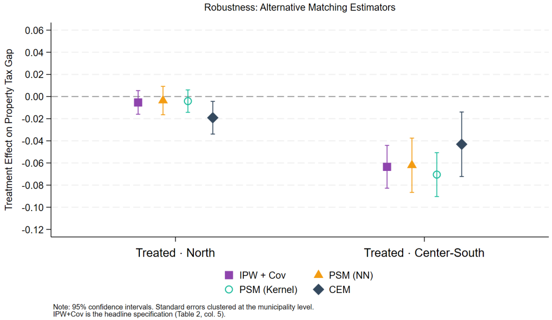

ATT across IPW+Cov (baseline), PSM-NN, PSM-Kernel, CEM. 95% CIs.

- PSM-NN (1:1, no repl.): CS −6.2 pp***
- PSM-Kernel (Epanechnikov): CS −7.1 pp***
- CEM (coarsened exact): CS −4.3 pp***

CS effect: −4.3 **to** −7.1 **pp***** across all four estimators.

North: null in three of four; CEM shows a small −1.9 pp** (subsample variance).

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